

Axion Strings AMR

Physics and Numerical Conventions

Network evolution only — see the code for loop/collision test ICs

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Scope. This is a physics reference, not a how-to: it specifies what equations are solved, how the field is initialised and relaxed, how refinement is decided, and how every reported observable is defined — the detail needed to interpret a run's output correctly, not to launch one. For parameter names, defaults and cluster mechanics, see the getting-started guide. Only the physical-string network pipeline is covered (`ic_mode = fourier_relaxed` into the main evolution); the flat-space loop/collision test ICs are a separate, simpler code path and are not described here.

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1 Field content, units, and normalisation

The field is a single complex scalar ϕ (the Peccei–Quinn field), written as two real components. Code units fix

$$v = 1, \quad \lambda_0 = 1, \quad \tau_0 = 1,$$

where v is the vacuum expectation value ($\langle |\phi| \rangle = v$), λ_0 the quartic coupling at τ_0 , and τ is *conformal* time throughout. The field is evolved in a rescaled, comoving form

$$\psi = R(\tau) \phi / v, \quad \Pi \equiv \psi' = d\psi / d\tau,$$

with $\psi = \psi_1 + i\psi_2$; the four state variables are $(\psi_1, \psi_2, \Pi_1, \Pi_2)$. Because $R(\tau) > 0$ is real, $\arg(\psi) = \arg(\phi)$ exactly — the phase is unaffected by the comoving rescaling, which is used throughout (string finding, masking) to work directly with the stored state.

Normalisation. ϕ is canonically normalised as a *complex* scalar: the kinetic and gradient terms carry no extra factor of $\frac{1}{2}$ (unlike a single real scalar). This was re-derived directly from the equation of motion (Section 2) rather than assumed, after finding an error of this kind in an earlier document — see `Energy.hpp`’s header for the check. The axion (phase) field is $a = f_a \theta$, $\theta = \arg(\phi)$, with

$$f_a = \sqrt{2} v = \sqrt{2} \quad \text{in code units.}$$

2 Background and equations of motion

2.1 Expansion law

The scale factor is a pure power law in conformal time,

$$R(\tau) = R_0 \left(\frac{\tau}{\tau_0} \right)^{1/b_{\text{inv}}}, \quad b_{\text{inv}} \equiv a_{\text{inv}} - 1, \quad R_0 = \frac{1}{b_{\text{inv}}},$$

so the single input `a_inv` fixes the whole expansion history (radiation domination is $a_{\text{inv}} = 2$). The physical Hubble rate is $H(\tau) = R'(\tau)/R(\tau)^2$, independent of everything below — only $\lambda(\tau)$, not $R(\tau)$, responds to the mass schedule.

2.2 Mass schedule and the three regimes

$$\lambda(\tau) = \lambda_0 \left(\frac{R(\tau)}{R_0} \right)^{-2c}, \quad m_r(\tau) = \sqrt{\lambda(\tau)}.$$

A single exponent `c0` (with an optional single switch to `c1` at a chosen τ_{sw} , λ re-anchored there so it stays continuous, only its logarithmic slope changes) selects which of three physically distinct schedules is being run:

- **Physical strings**, $c = 0$: $\lambda \equiv \lambda_0$ is exactly constant, so the string’s comoving core width is fixed while the comoving horizon grows — m_r/H grows with τ , which is the actual target dynamics (real global strings), and is why $\log(m_r/H)$ is the natural “how far into the run” coordinate used everywhere in this project.
- **Fat strings**, $c = 1$: $\lambda \propto R^{-2}$, chosen so that $R(\tau) m_r(\tau)$ is exactly constant — the comoving core width shrinks at exactly the rate the comoving horizon does, holding m_r/H *fixed*. Cores are artificially fat/cheap to resolve; used to relax a network onto its attractor fast (Section 4) and, via a switch to $c = a_{\text{inv}}$ mid-run, as a bridge into Moore’s trick.

- **Moore's trick**, $c = a_{\text{inv}}$: singular in the general box-planning formula below (the exponent $(a_{\text{inv}} - 1)/(a_{\text{inv}} - c)$ diverges), handled by its own formulas (Section 3). HL (Hubble length over box size) falls as $1/\tau$ while resolution only improves, letting a single run reach a much larger $\log(m_r/H)$ than the physical schedule could afford at the same cost – supporting evidence for the main physical-string result, not the headline measurement itself.

2.3 Equation of motion

The complex-scalar equation of motion actually integrated (one RHS, templated on nothing — physical, fat and Moore differ *only* through $\lambda(\tau)$ above, never through separate code paths) is

$$\Pi'_i = \nabla^2 \psi_i + c_{\text{curv}}(\tau) \psi_i - \frac{\lambda(\tau)}{2} \psi_i (|\psi|^2 - R(\tau)^2), \quad \psi'_i = \Pi_i, \quad i = 1, 2,$$

with the Laplacian a comoving spatial derivative (fourth-order stencil) and

$$c_{\text{curv}}(\tau) = \frac{1 - b_{\text{inv}}}{b_{\text{inv}}^2 \tau^2}.$$

$R(\tau)$, $\lambda(\tau)$ and $c_{\text{curv}}(\tau)$ are evaluated on the host from τ and passed into the kernel as plain numbers – no parameter parsing or background evaluation inside the per-cell loop, so the kernel is GPU-clean by construction.

No Kreiss–Oliger dissipation. $\sigma = 0$ is the standing default: KO dissipation damps precisely the high- k modes the axion spectral index q is measured from, so it is not added as a matter of course. A nonzero σ , if ever required for stability, must be characterised (its effect on the spectrum quantified) with the same care as the lattice spacing, not quietly switched on.

3 Box planning

Two physical targets, both evaluated at the run's final time τ_f , fix the comoving box:

$$\begin{aligned} N_1 &= H(\tau_f) \tilde{L}(\tau_f) && \text{(Hubble patches per box length),} \\ N_2 &= \frac{1}{\Delta(\tau_f) m_r(\tau_f)} && \text{(grid points per core width),} \end{aligned}$$

with $N = N_1 N_2$ (comoving box length in finest-grid cells) the third input. For $c \neq a_{\text{inv}}$,

$$\tau_f = \left(\frac{N}{N_1 N_2} \right)^{\frac{a_{\text{inv}} - 1}{a_{\text{inv}} - c}}, \quad \tilde{L} = \frac{N_1}{R_0} \left(\frac{N}{N_1 N_2} \right)^{\frac{a_{\text{inv}} - 1}{a_{\text{inv}} - c}}.$$

For AMR, N/N_2 above refer to the *finest* level; the base (level-0) grid is $N/2^{\text{max_level}}$. Each refinement level is only permitted once the current finest level's own resolution has genuinely degraded to the N_2 target – the level-addition schedule, spaced by

$$\Delta \log(m_r/H) \text{ per level} = \ln 2 \cdot \frac{a_{\text{inv}} - c}{1 - c}$$

(singular at $c = 1$: a fat run's comoving core width never shrinks, so no further level is ever needed once set up). Buffering between regrid (`amr.n_error_buf`) is sized so that the fastest possible signal ($v = 1$ in these units) cannot cross out of the tagged-plus-buffer region before the next regrid check, given the fixed relation $dt_{\text{level}} = dt_{\text{multiplier}} \times dx_{\text{level}}$.

4 Initial conditions and pre-evolution

4.1 Why relax at all

Starting the physical-string ($c = 0$) schedule directly from a raw random field would spend a large part of the run just forming a realistic tangle, and the relaxation transient itself contaminates later diagnostics. Instead, an over-seeded Gaussian random field is generated and evolved under a separate, favourable schedule until it settles onto the network's own attractor, *then* handed off into the main ($c = 0$) run.

4.2 Fourier-mode initial field

ψ_1, ψ_2 are drawn as an independent Gaussian random field per Fourier mode (deterministic per-mode hash of (seed, component, i, j, k), not a stream-based RNG – reproducible at fixed (seed, parameters, rank count)), flat over $|k| \leq k_{\max}$ and zero above, with

$$k_{\max} \text{ (in cells)} = k_{\max_over_mr} \times m_r(0) \times dx,$$

then rescaled in real space to hit the requested mean-square variance exactly. $k_{\max}$ is deliberately *not* restricted to the core scale: sub-core modes seed extra, doomed-to-annihilate vortex–antivortex pairs (Kibble mechanism), and since annihilation only ever removes net winding, under-seeding can never reach a dense target network – $k_{\max}$ must be generous enough that the network is over-tangled at $\tau_{\text{pre}} = 0$ and has genuine room to relax down.

4.3 The pre-evolution background

Relaxation runs on its own clock τ_{pre} (starting at 0) and its own background, *not* an instance of Section 2's power law ($a_{\text{inv}} = 1$ is singular there). Switching to τ_{pre} turns a linear-in-cosmic-time R into

$$R_{\text{pre}}(\tau_{\text{pre}}) = e^{\tau_{\text{pre}}}, \quad \lambda_{\text{pre}}(\tau_{\text{pre}}) = \frac{\gamma_{\text{pre}}^2}{R_{\text{pre}}(\tau_{\text{pre}})^2}, \quad c_{\text{curv}}^{\text{pre}} = 1 \text{ (constant)},$$

($R_0 = \alpha = 1$ WLOG, one physical input γ_{pre} remaining). This is exactly the fat-mode $\lambda \propto R^{-2}$ shape (Section 2.2) under this new time gauge, so it reuses the same RHS unchanged, and by construction holds

$$\left. \frac{m_r}{H} \right|_{\text{pre}} = \gamma_{\text{pre}} \quad \text{exactly constant throughout relaxation,}$$

letting the network converge onto a genuine attractor fast rather than chasing a moving target. Unless set explicitly, $\gamma_{\text{pre}} = 1/dx_{\text{base}}$ (resolves the core with ~ 1 cell at $\tau_{\text{pre}} = 0$) – purely a starting-resolution choice; because λ_{pre} keeps decreasing through relaxation, the comoving core width *grows* over the course of relaxation, away from that starting point.

4.4 Stopping condition

Relaxation stops the first time the measured ξ (Section 7.1, on the relaxation's own grid, at an adaptive check cadence that tightens as ξ approaches target) drops to `xi_target`. ξ generally decreases monotonically as the over-seeded network anneals.

4.5 Handoff to the main run

At the stopping instant $\tau_{\text{pre,end}}$, the field is rescaled into the main run’s own normalisation. Writing $\phi = \psi/R$ in either frame and requiring ϕ itself to be continuous across the switch gives, with $\kappa \equiv R_{\text{main}}(\tau_i)/R_{\text{pre}}(\tau_{\text{pre,end}})$,

$$\psi_i \rightarrow \kappa \psi_i, \quad \Pi_i \rightarrow \kappa^2 \Pi_i + \frac{\psi_i}{R_{\text{pre}}} \left(R'_{\text{main}}(\tau_i) - \kappa^2 R'_{\text{pre}}(\tau_{\text{pre,end}}) \right),$$

an exact chain-rule identity (not an approximation): it matches ψ and $d\psi/d\tau$ under the new background with no assumption beyond ϕ ’s own continuity. The main run’s own clock then begins at τ_i (`log_mr_over_h_i`), continuing from the same integrator state – no restart or checkpoint round-trip.

4.6 Handoff smoothing

The rescale above is exact for ψ, Π , but λ and c_{curv} themselves still jump discontinuously at the handoff (the pre-evolution and main schedules are, in general, different curves in $\lambda(\tau)$ that need not agree at whatever instant relaxation happens to stop). A discontinuous jump in λ is a discontinuous jump in the string core’s natural width, and leaving the field’s already-formed spatial profile as-is while its equation of motion suddenly wants a different equilibrium width excites a measurable transient: an oscillatory exchange between radial kinetic and potential energy, and a bump in the axion spectrum near the old core scale that migrates towards the new one over several τ . By default (`handoff_transition_n_periods = 3`), λ and c_{curv} are instead blended from their frozen pre-evolution values to the main schedule’s own values via a smoothstep ($3t^2 - 2t^3$, value-continuous with zero slope at both ends, so the smoothing itself adds no new discontinuity) over this many main-schedule core-oscillation periods, $2\pi/\sqrt{\lambda_{\text{main}}(\tau_i)}$. Set to 0 to recover the instantaneous jump.

5 Tagging and AMR refinement

The primary refinement criterion is exact and parameter-free: a cell is tagged if any of its three low-index-corner plaquettes (xy, yz, zx) is pierced, i.e. the phase $\theta = \arg(\psi)$ winds by a full $\pm 2\pi$ around the plaquette’s four corners (each consecutive step wrapped into $(-\pi, \pi]$ before summing). This is the *same* routine the ξ diagnostic uses to count strings (Section 7.1) – one implementation, not two copies that could silently drift apart. A level may only be created once $\log(m_r/H)$ has actually crossed its own threshold (Section 3); no refinement happens during relaxation at all (relaxation’s own clock is not $\log(m_r/H)$ in the main schedule’s sense, and holding m_r/H fixed throughout relaxation means no new level is ever due there regardless).

Two further, optional per-cell triggers exist – a Laplacian threshold on ψ_1, ψ_2 (`gradient_threshold`) and a radial-amplitude-gradient threshold (`radial_gradient_threshold`) – both *off by default*: calibrating either against an isolated static core was tried and found to flag essentially the whole domain once applied to a real, dense network’s field statistics, not just cores, so neither is trusted as a default.

Refinement-systematics control. `force_full_refinement` (off by default) bypasses both the criteria above and the schedule-gating, tagging every cell at every level unconditionally – not a production setting, but a way to run an otherwise-identical pair of simulations, one adaptively refined and one uniformly refined to the same finest resolution, to check directly whether refinement itself is biasing any observable.

6 Masking (screening)

The axion phase rate $\theta' = (\psi_1 \Pi_2 - \Pi_1 \psi_2)/|\psi|^2$ genuinely diverges near a string core ($|\psi| \rightarrow 0$), so any spectrum or energy built from it must exclude the core region before transforming or summing – applied at exactly one place (filling the buffer handed to the FFT/reduction), not inlined at several call sites, so an index or threshold error cannot silently diverge between them. Three schemes:

- **None** ($f = 1$): unscreened pass, required for every screened/unscreened energy or spectrum pair (Sections 7.2, 7.4) and for the tension’s core+tail term.
- **A**: the bare numerator $\psi_1 \Pi_2 - \Pi_1 \psi_2$, no division, no top-hat.
- **B**: the genuine phase rate θ' above, top-hat zeroed wherever $|\psi|/R$ is below a runtime threshold (default 0.8, not a settled choice – 0.9, 0.95 have also been tried without reconciling on one value). The same 0/1 top-hat is also the weight used to correct the effective point count in any spatial average over masked data (e.g. n_{unmasked} below).

7 Observables

7.1 String density ξ

Each pierced plaquette represents $\sim \frac{2}{3} dx$ of string length (a *statistical* correction for random orientation relative to the lattice – exact for a network of many randomly-oriented strings, not for a single deterministically axis-aligned one). Summing across AMR levels by comoving length (never by raw cell/plaquette count, which would overweight finer levels),

$$\ell_{\text{comoving}} = \frac{2}{3} \sum_{\ell} N_{p,\ell} d\mathbf{x}_{\ell}, \quad \xi = \frac{\ell_{\text{comoving}}}{\tilde{L}^3} \cdot \frac{(a_{\text{inv}} - 1)^3}{a_{\text{inv}}^2} \cdot \tau^2.$$

N_p (“plain”) counts pierced plaquettes; $N_{p,\text{weighted}}$ sums $|\text{winding}|$ over them – most strings are singly wound, so the two agree closely, and a divergence between them is itself a diagnostic.

7.2 Energies

The regular (non-singular) total physical energy density, from the canonically normalised ϕ Lagrangian directly (not the polar axion+radial decomposition below, which is individually singular at the core):

$$\rho_{\text{tot}} = \frac{1}{R^4} \left[|\Pi - (R'/R)\psi|^2 + |\nabla\psi|^2 + \frac{1}{4}\lambda(|\psi|^2 - R^2)^2 \right].$$

The phase (axion) and amplitude (radial/Higgs) modes are also reported individually, each the genuine energy carried by that one degree of freedom alone (not “total minus the other”):

$$\rho_a^{\text{kin}} = \frac{1}{2}\dot{a}^2, \quad \rho_a^{\text{grad}} = \frac{1}{2}(\nabla a)^2, \quad a = f_a \theta,$$

with $\dot{a} = f_a \theta'/R$ (cosmic time from conformal) and ∇a picking up one power of $1/R$ (physical), not two (a pure phase has no amplitude to redshift). The radial energies use the exact orthogonal decomposition of $|\dot{\phi}|^2/|\nabla\phi|^2$ into radial and tangential parts (a Pythagorean identity, not an approximation): $r \equiv |\phi| - v$,

$$\rho_r^{\text{kin}} = \dot{r}^2, \quad \rho_r^{\text{grad}} = (\nabla r)^2, \quad \rho_r^{\text{mass}} = \frac{1}{4}\lambda(|\phi|^2 - v^2)^2.$$

Every energy is reported both screened and unscreened (Section 6).

7.3 String tension

Two variants, both normalised by ℓ_{comoving} above, from the composite (all-level) coverage sums $n_{\text{total}}/n_{\text{unmasked}}$ (comoving *volumes* once levels are combined, not raw counts):

$$\mu_{\text{core}} = \frac{R^2 [n_{\text{total}} \rho_{\text{tot}}^{\text{unscr}} - n_{\text{unmasked}} \rho_{\text{tot}}^{\text{scr}}]}{\ell_{\text{comoving}}}, \quad \mu_{\text{core+tail}} = \mu_{\text{core}} + \frac{R^2 n_{\text{unmasked}} [\rho_{a,\text{scr}}^{\text{grad}} - \rho_{a,\text{scr}}^{\text{kin}}]}{\ell_{\text{comoving}}}.$$

μ_{core} is expected to converge to a resolution-independent *constant*, but has no simple closed form: it depends on both the near-core profile (not universal) and the masking threshold's exact radius. $\mu_{\text{core+tail}}$ is expected to *diverge logarithmically* with the box/horizon size – the long-range Goldstone tail is universal, and its coefficient is calculable directly from the axion gradient energy density above: at large ρ from an isolated winding-1 core, $|\nabla\theta|^2 = 1/\rho^2$, so

$$\rho_a^{\text{grad}}(\rho) = \frac{1}{2} f_a^2 / \rho^2 = 1/\rho^2 \quad (v = 1) \quad \implies \quad \mu_{\text{tail}} = 2\pi \int_{1/m_r}^L \rho \frac{d\rho}{\rho^2} = 2\pi \ln(m_r L).$$

Taking $L \sim 1/H$ (the natural IR cutoff for a single string in an expanding background) gives the closed-form prediction

$$\mu_{\text{core+tail}} \approx 2\pi \ln(m_r/H)$$

directly in terms of the reported m_r/H column. This is the *isolated single-string* estimate; a real network's nearby strings partially screen each other's long-range tail, so a measured value somewhat below this is expected, not a discrepancy.

7.4 Spectrum

The masked $a_{\text{dot}} = f_a \theta' / R$ field (Section 6) is Fourier-transformed (`amrex::FFT::R2C`, level-0 only – established policy, not yet extended across the hierarchy) and binned into shells by integer mode number $p = \lfloor |\mathbf{p}| + 0.5 \rfloor$, dropping modes outside the inscribed sphere ($p \geq N/2$). Reported alongside the bare shell average and mode index is the physical k/H for that mode,

$$k/H = \frac{2\pi p}{\tilde{L} H(\tau)},$$

the x -axis a spectral index q is actually fit against. Screened and unscreened spectra are both saved, from two independent FFT passes (scheme B and None), so the impact of screening itself is directly visible.

7.5 String velocities

Evaluated at the corners of every pierced plaquette (global-string equilibrium profile coefficients $c_1 = 0.41222$, $e_1 = -0.025763$), R-independent by construction (a physical velocity cannot depend on the comoving rescaling used to evolve the field – verified as an exact algebraic cancellation, not assumed):

$$\gamma^2 v^2 = \frac{|\dot{\phi}|^2}{2m_r^2 c_1^2} \left(1 - \frac{e_1 |\phi|^2}{c_1^3} \right) - \frac{e_1 [\text{Re}(\phi^* \dot{\phi})]^2}{4m_r^2 c_1^5}.$$